KINEMATICS

Distance and Displacement

Distance

Distance is simply the total path length traversed in moving from one position to the other. In general, the distance between two points depends on the path traveled. Distance is a scalar quantity.

Displacement

Displacement is defined as a straight line distance between two points, along the direction from the starting point to the final point. Displacement is a vector quantity.

If a particle moves from position vector r_1 to r_2

Where $r_1 = x_1\boldsymbol{i} + y_1\mathbf{j} + z_1\mathbf{k}$

$r_2 = x_2\boldsymbol{i} + y_2\mathbf{j} + z_2\mathbf{k}$

Then the displacement vector is

$\Delta r = r_2 - r_1 = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$

$\Delta r = \Delta x\mathbf{i} + \Delta y\mathbf{j} + \Delta z\mathbf{k}$

Speed and Velocity

Speed

When a body is in motion, its position changes with time i.e., it moves a certain distance in a given time. This implies that both length and time are important quantities when describing motion. The rate at which distance covered is speed. Average speed is the total distance *d* traveled, divided by the total time *t* elapsed in traveling that distance

$$Average\ speed = \frac{total\ distance}{total\ time} = \frac{d}{t} \qquad \textbf{4.1}$$

Speed is a scalar quantity.

Velocity

Velocity is a vector quantity used to describe motion. Velocity is defined as rate of change of displacement with time. We have the average velocity defined as displacement divided by time taken:

$$\text{Average velocity} = \frac{displacement}{change\ \text{in}\ time} = \frac{\Delta \mathbf{x}}{\Delta t} = \frac{x - x_0}{t - t_0} = \bar{v} \qquad \textbf{4.2}$$

Where x_o is the initial point, x is final position, t_o is initial time, t is final time, $\Delta \boldsymbol{x}$ is the change in position or displacement and Δt is the change in t.

The SI unit of velocity is meters per second (m/s). Instantaneous velocity on the other hand is defined as

$$\text{Instantaneous velocity} = \frac{d\mathbf{x}}{dt} = \lim_{t \to 0} \frac{x - x_0}{t - t_0} \qquad \textbf{4.3}$$

Acceleration

Average acceleration is defined as *change in velocity divided by time taken*:

$$\text{Average acceleration} = \frac{change\ \text{in}\ velocity}{change\ \text{in}\ time} = \frac{\Delta v}{\Delta t} = \frac{v - v_0}{t - t_0} = \bar{a} \qquad \textbf{4.4}$$

Accelerated Motion

Since velocity is a vector quantity with both magnitude and direction, an acceleration can occur when there is a

(a) change in magnitude of the velocity but not the direction

(b) change in the direction of the velocity but not the magnitude

(c) change in both the magnitude and direction

Example 4.1 Mr. Tetteh and his driver are traveling to Cape Coast in a Toyota pickup which is moving at 90 km/h down a straight highway. The driver of the pickup sees an accident in the distance and slows down to 40 km/h in 5.0 s. What is the average acceleration of the Toyota pickup?

Solution

Given the initial and final velocities and the time duration

$v_0 = (90km/h) = (25m/s)$, $v = (40km/h) = (11m/s)$, $t = 5.0s$

NB: converting from km/h to m/s

$$\frac{90\ x1000}{60\ x\ 60} = \frac{900}{36} = 25\ m/s,$$

$$\frac{40\ x1000}{60\ x\ 60} = \frac{400}{36} = 11.11\ m/s$$

We are solving to find: average acceleration, $\bar{a}$, which from Equation 4.4 is

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v - v_0}{\Delta t} = \frac{11.11m/s - 25m/s}{5.0s} = -2.8m/s$$

Average Velocity and Instantaneous velocity in components

To know how fast a particle moves from one point to the other, you have to consider

1. Average velocity and
2. Instantaneous velocity

These are vectors and so vector notation has to be used.

If a particle moves through a displacement $\Delta\mathbf{r}$ in a time interval Δt, then its average velocity $\bar{v}$ is

$$\bar{v} = \frac{\text{displacement}}{\text{time taken}} = \frac{\Delta\mathbf{r}}{\Delta t} \qquad \textbf{4.5}$$

This tells us that, the direction of $\bar{v}$ is the same as that of $\Delta\mathbf{r}$. $\bar{v}$ can also be written in unit vector form as

$$\bar{v} = \frac{\Delta x\mathbf{i} + \Delta x\mathbf{j} + \Delta x\mathbf{k}}{\Delta t} = \frac{\Delta x}{\Delta t}\mathbf{i} + \frac{\Delta y}{\Delta t}\mathbf{j}\frac{\Delta z}{\Delta t}\mathbf{k} \qquad \textbf{4.6}$$

The velocity of a particle usually refers to the instantaneous velocity $\mathbf{v}$ which is the value that $\bar{v}$ approaches in the limit as we shrink the time interval to 0 about the instant. Using the language of calculus, we may write $\mathbf{v}$ as the derivative

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} \qquad \textbf{4.7}$$

The direction of the instantaneous velocity $\mathbf{v}$ of a particle is always tangent to the particles path at the particles position.

In substituting $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ into Equation 4.7 we get

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k} \qquad \textbf{4.8}$$

$$\mathbf{v} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k} \qquad \textbf{4.9}$$

where, $v_x = \frac{dx}{dt}, v_y = \frac{dy}{dt}, v_z = \frac{dz}{dt}$

. Since $v_x = \frac{dx}{dt}, v_y = \frac{dy}{dt}, v_z = \frac{dz}{dt}$ are the scalar components of $\mathbf{v}$ along the x, y and z -directions, it means we can find the scalar components of $\mathbf{v}$ by differentiating the scalar components of $\mathbf{r}$.

Average acceleration and instantaneous acceleration in components

As a particle's velocity changes from $\mathbf{v}_1$ to $\mathbf{v}_2$ in a time interval Δt, its average acceleration $\bar{a}$ during Δt is

$$\bar{a} = \frac{\text{change in velocity}}{\text{time interval}} = \frac{\mathrm{v}_2 - \mathrm{v}_1}{\Delta \mathrm{t}} = \frac{\Delta \mathrm{v}}{\Delta \mathrm{t}} \qquad \textbf{4.10}$$

If Δt is zero about some instant, then in the limit approaches the instantaneous acceleration (or acceleration $\bar{a}$) at the instant; that is

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}) = \frac{dv_x}{dt}\mathbf{i} + \frac{dv_y}{dt}\mathbf{j} + \frac{dv_z}{dt}\mathbf{k} \qquad \textbf{4.11}$$

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k} \qquad \textbf{4.12}$$

where $a_x = \frac{dv_x}{dt}$, $a_y = \frac{dv_y}{dt}$ and $a_z = \frac{dv_z}{dt}$. We can therefore find the scalar components of $\mathbf{a}$ by differentiating the scalar components of $\mathbf{v}$.

Example 4.2 If a rabbit's velocity is $v = -0.62t^2\mathbf{i} + 7.2\mathbf{j}$, find the acceleration $\mathbf{a}$ at the time $t = 5s$, in unit-vector notation and in magnitude notation.

Solution

Given that $v = -0.62t^2\mathrm{i} + 7.2\mathbf{j}\ m/s$

$$a = \frac{dv}{dt} = -0.62(2t)i = -1.24t\mathrm{i}$$

At time $t=5s$, $a = \frac{dv}{dt} = -1.24(5)i = -6.2i\ \mathrm{ms}^{-2}$

$$|a| = \left|\frac{dv}{dt}\right| = |-1.24(5)i| = \sqrt{(-6.2)^2} = 6.2\ \mathrm{ms}^{-2}$$

Kinematic equations and problem solving

Any rectilinear motion can be defined by the position of that body in space at any given instant. In order to make such a definition to describe a motion, we need to find the relationships between distance and displacement, speed and velocity, acceleration and time. The kinematic equations are equations of motion that relate the above-mentioned quantities with each other. The assumption of a uniform motion is made in all cases for use of these kinematics equations of motion. The following kinematics equations will not work, for example, for an object that is accelerating erratically. Below is the list of the kinematic equations.

Average velocity, $\mathbf{v} = \frac{\mathbf{s}}{t}$	**4.13**
Displacement, $\mathbf{s} = \frac{1}{2}(\mathbf{v} - \mathbf{u})t$	**4.14**
Average acceleration, $\mathbf{a} = \frac{\Delta\mathbf{v}}{\Delta t} = \frac{\mathbf{v} - \mathbf{u}}{\Delta t}$	**4.15**
$\mathbf{v} = \mathbf{u} + \mathbf{a}t$	**4.16**
$\mathbf{s} = \mathbf{x} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$	**4.17**
$v^2 = u^2 + 2as$	**4.18**

Note that in some of the equations above we have replaced the time interval Δt with t, representing total time.

Example 4.3 A car traveling at a speed of +12 m/s accelerates at a rate of +4 m/s^2 for a time of 5s. What is the car's speed at the end of 5s?

Solution

Given u = 12m/s, a = 4m/s, t = 5s, v = ?. We use Equation 4.16 to find v

$$v = u + at = 12m/s + (4m/s^2 \times 5s) = 12m/s + 20m/s = 32m/s$$

Derivation of the equations of motion

Suppose that an object is initially moving with a constant acceleration a, and increases its velocity from u to v and its displacement from 0 to s in the time interval t, then since acceleration is the rate of change of velocity, we have

$$a = \frac{v-u}{t}$$

Rearranging gives

$$v = u + at. \qquad \textbf{4.16}$$

The average velocity is $\frac{1}{2}(u+v)$ and therefore since displacement is $average\ velocity \times time$, we have

$$s = \frac{1}{2}(v-u)t \qquad \textbf{4.14}$$

But from Equation 4.16, we can write

$$s = \frac{1}{2}(u+u+at)t = ut + \frac{1}{2}at^2 \qquad \textbf{4.17}$$

From Equation 4.15 $t = (v-u)/a$ and substituting into Equation 4.17 we have

$$s = (\frac{v+u}{2})(\frac{v-u}{a})$$

$$v^2 - u^2 = 2as$$

$$\therefore v^2 = u^2 + 2as \qquad \textbf{4.18}$$

Graphs of Motion

The motion of a body or an object in a straight line (rectilinear motion) can be examined by graphical analysis. Graphical analysis is often helpful in understanding motion and its related quantities. For example the motion of a runner can be graphed as a plot of position x versus time t.

We have

(a) displacement-time graph

(b) velocity-time graph and

(c) acceleration-time graph

Displacement-Time Graph

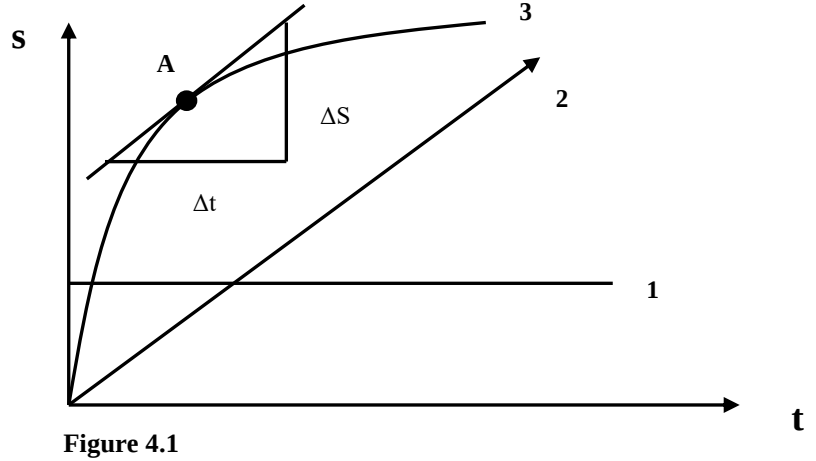


Figure 4.1

From Figure 4.1 we can make the following deductions about the following curves:

(1) slope = 0: velocity is zero (i.e. the body is stationary)

(2) slope = constant : The body moves with uniform velocity

(3) The slope varies at different points and the gradient at a given point say A is the instantaneous velocity. This implies the body moves with a non-uniform velocity.

Velocity-Time Graph

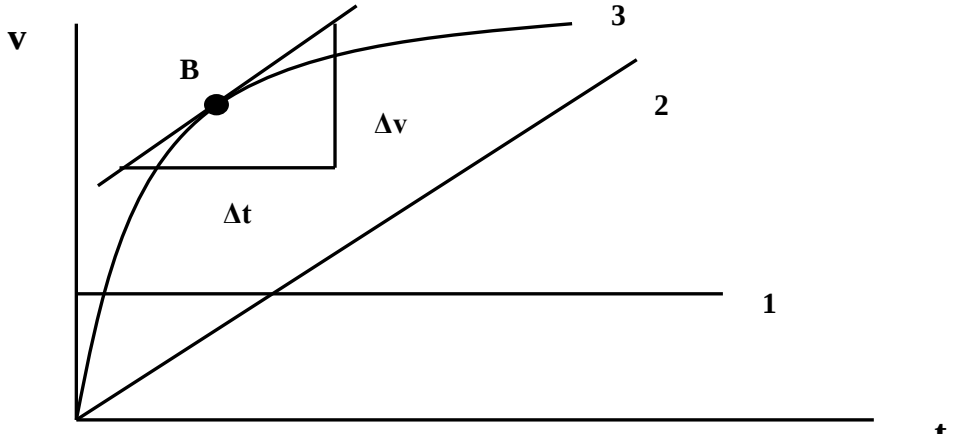


Figure 4.2

From figure4.2:

(1) slope = 0: acceleration is zero and hence velocity is constant

(2) slope = constant: The body moves with uniform acceleration

(3) The slope varies at different points and the gradient at a given point say B is the instantaneous acceleration. The body moves with a non-uniform acceleration. Note that for Velocity-Time graph the area under the graph is equal to displacement.

Acceleration-Time Graph

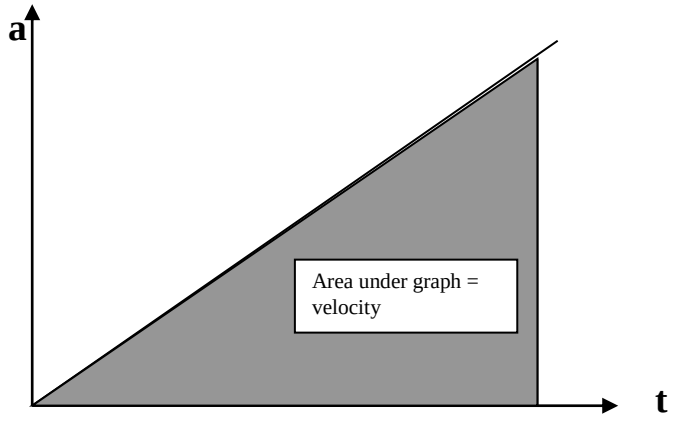


Figure 4.3

For acceleration-Time graph the area under the graph is equal to the velocity.

Free fall (motion under gravity) and acceleration due to gravity

An object is in free fall if it falls freely (i.e. in conditions similar to a vacuum without any resistance to motion) under the influence of gravity. Any object, regardless of its mass which falls freely near the surface of earth falls with an average acceleration of 9.8 m/s^2. Since the free fall acceleration is constant at a given point, we can use the kinematic equations of motion to solve problems involving free fall.

NB: using equations 4.16, 4.17 and 4.18, acceleration a = g (downward motion) and a = -g (upward motion). Distance s = height h.

Example 4.4 A ball is dropped from rest from a high window of a tall building and falls for 3s. Neglecting air resistance (a) what is the speed of the ball? (b) What is the distance the ball has fallen from where it was released? (c) What is the

ball's acceleration? [Assume that g = 10 m/s]. What distance will it travel in the last second (i.e. between t = 2 and t = 3)?

Solution

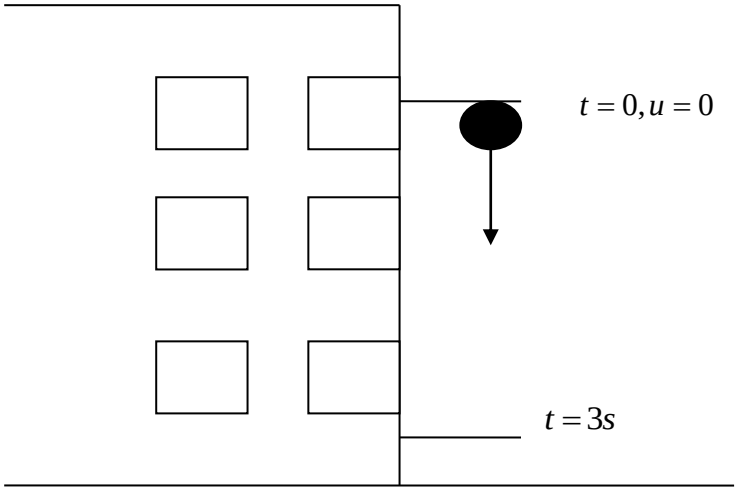


Figure 4.4

We start solving by listing the quantities already given in this problem

$u = 0$ since the ball was dropped from rest, $t = 3s$, $a = g = 10m/s^2$ since the ball is falling freely under the influence of gravity.

(a) We are looking for the speed *v* after 3s, so we find an equation that includes *v*, *u*, *a* and t. Equation 4.16 gives $v = u + gt = 0 + (10m/s^2)(3s) = 30m/s$

(b) The distance fallen can be found by using Equation 4.17, choosing the downward direction as positive:

$$s = ut + \tfrac{1}{2}gt^2 = (0)(3s) + \tfrac{1}{2}(10m/s^2)(3s)^2 = 45m$$

Or use $\dfrac{v^2 - u^2}{2a} = s$

(c) After 3s of free fall, the ball is still accelerating at 10m/s^2. The ball's speed is increasing, but the rate at which its speed is increasing (i.e. its acceleration) is constant.

$$S_2 = \frac{1}{2}.10.2^2; \qquad S_3 = \frac{1}{2}.10.3^2$$

$$S_2 = 20, \quad S_3 = 45; \quad S_3 - S_2 = 45 - 20 = 25\text{m}$$

Relative Motion and Velocity in one dimension

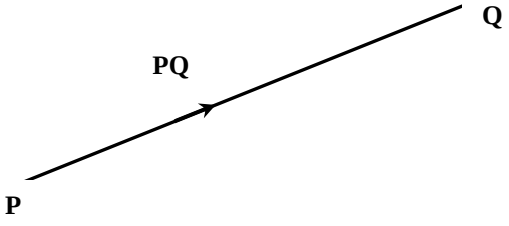


Figure 4.5

Suppose that P and Q are moving points. Then **PQ** represents the position vector of Q relative to P, and $\frac{d}{dt}(\mathbf{PQ})$ represents its rate of change, which we shall call the velocity of Q relative to P and is written as $\mathbf{V}_{PQ}$. So

$$\mathbf{v}_{PQ} = \frac{d}{dt}(\mathbf{PQ}) \qquad \textbf{4.19}$$

If we consider three moving points P, Q and R

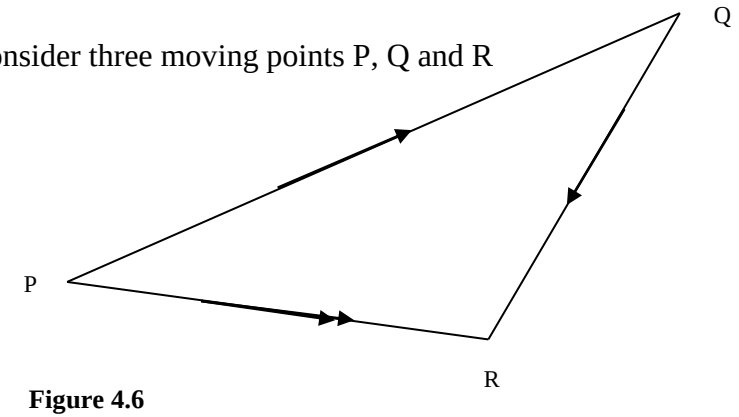


Figure 4.6

$$\mathbf{PQ} + \mathbf{QR} = \mathbf{PR} \qquad \textbf{4.20}$$

Hence differentiating with respect to time we have

$$\frac{d}{dt}(\mathbf{PQ}) + \frac{d}{dt}(\mathbf{QR}) = \frac{d}{dt}(\mathbf{PQ} + \mathbf{QR}) = \frac{d}{dt}(\mathbf{PR}) \qquad \textbf{4.21}$$

$$\Rightarrow \mathbf{v}_{PQ} + \mathbf{v}_{QR} = \mathbf{v}_{PR} \qquad \textbf{4.22}$$

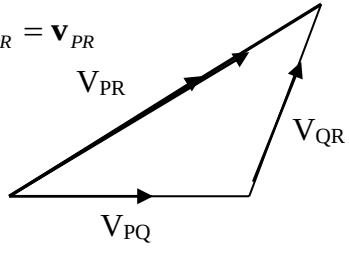


Figure 4.7

Furthermore, the rate of change of $\mathbf{v}_{PQ}$ can be called the acceleration of Q relative to P and is written as

$$\mathbf{a}_{PQ} = \frac{d}{dt}(\mathbf{v}_{PQ}) \qquad \textbf{4.23}$$

and it immediately follows that

$$\mathbf{a}_{PQ} + \mathbf{a}_{QR} = \mathbf{a}_{PR} \qquad \textbf{4.2}$$

Circular Motion

Uniform Circular Motion

A particle is in uniform circular motion if it travels around a circle or a circular arc at constant speed. A car moving in a circular path with linear constant speed v is in uniform circular motion. Although the speed does not vary, the car is accelerating. Since the car's direction of motion changes, the car has acceleration. This fact may be surprising because acceleration is usually thought of as an increase or decrease in speed. But actually **v** is a vector, not a scalar. If **v** changes, even only in direction, there is acceleration, as happens in uniform circular motion. For any motion, the velocity vector is tangent to the path. Consequently, when an object moves in a circular path, its velocity vector is perpendicular to the radius of the circle.

To find the magnitude and direction of the acceleration, consider Figure 4.8a which shows a particle first at point A and then at point B. Let the particle be at point A at time t_i, with velocity $\mathbf{v}_i$ and at point B at time t_f with velocity $\mathbf{v}_f$. Assume that $\mathbf{v}_i$ and $\mathbf{v}_f$ have the same magnitudes (speeds) but differ only in direction.

We know from the definition of average acceleration that:

$$\bar{\mathbf{a}} = \frac{\mathbf{v}_f - \mathbf{v}_i}{t_f - t_i} = \frac{\Delta \mathbf{v}}{\Delta t} \qquad \textbf{4.25}$$

Consider the triangle in Figure 4.8b with sides $\Delta\mathbf{r}$ and $\mathbf{r}$. This triangle is similar with the one in Figure 4.8a having sides $\Delta\mathbf{v}$ and $\mathbf{v}$. From the two triangles we can write

$$\frac{\Delta v}{v} = \frac{\Delta r}{r} \qquad \textbf{4.26}$$

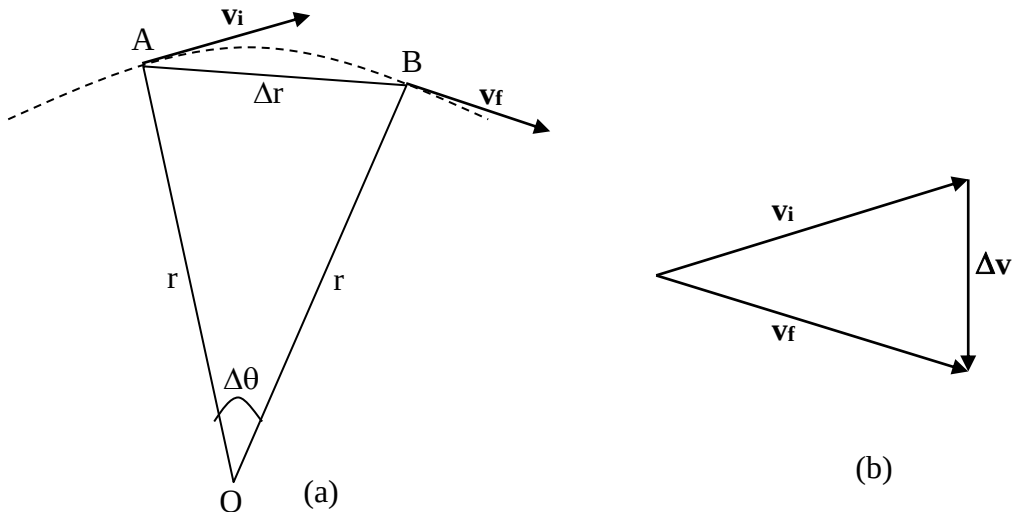


Figure 4.8 (a) As a particle moves from A to B, its velocity vector changes from $\mathbf{V_i}$ to $\mathbf{V_f}$. (b) The construction for determining the direction of the change in velocity $\mathbf{\Delta V}$, which is toward the center of the circle for small $\mathbf{\Delta r}$

Solve for $\Delta\mathbf{v}$ and substitute in the expression $\bar{\mathbf{a}} = \Delta\mathbf{v}/\Delta t$ to get

$$\bar{\mathbf{a}} = \frac{\mathbf{v}\Delta r}{r\Delta t} \qquad \textbf{4.27}$$

If points A and B are extremely close together, then $\mathbf{\Delta v}$ points toward the centre of the circular path, and because the acceleration is in the direction of $\mathbf{\Delta v}$, the acceleration also points toward the centre. In the limit as Δt approaches zero, the ratio $\Delta r/\Delta t$ also goes to the speed $\mathbf{v}$ and the magnitude of the acceleration is

$$a_r = \frac{v^2}{r} \qquad \textbf{4.28}$$

We then conclude that, when a particle moves at constant speed v in a circle of radius r, the acceleration of the particle is directed toward the centre of the circle and has a constant magnitude v^2/r.

Consider a particle moving along a curved path where the velocity changes in both direction and magnitude, as shown in Figure 4.9 As always, the velocity vector is tangent to the path, but the direction of the acceleration vector **a** changes from point to point. This vector can be resolved into two components: a radial component $\mathbf{a}_r$ and a tangential component $\mathbf{a}_t$. Thus **a** becomes

$$\mathbf{a} = \mathbf{a}_r + \mathbf{a}_t \qquad \textbf{4.29}$$

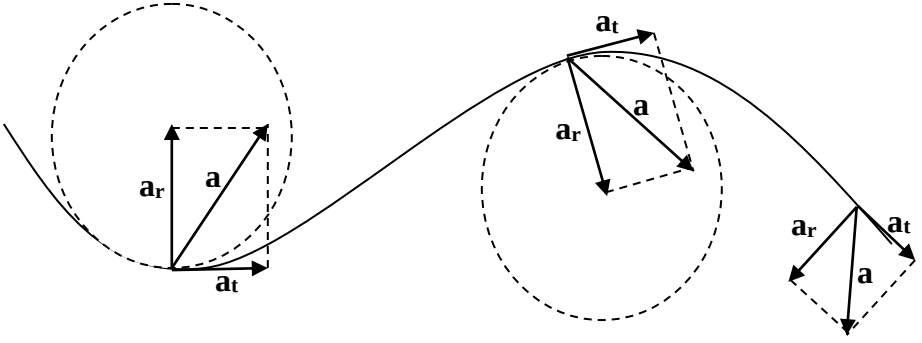


Figure 4.9 The motion of a particle along an arbitrary curved path lying in the xy plane. If the velocity vector **v** (always tangent to the path) changes in direction and magnitude, the component vectors of the acceleration **a** are a tangential component $\mathbf{a_t}$ and a radial component $\mathbf{a_r}$.

The tangential acceleration causes the change in the speed of the particle and it is parallel to the instantaneous velocity, and its magnitude is

$$a_t = \frac{d|\mathbf{v}|}{dt} \qquad \textbf{4.30}$$

The radial acceleration arises from the change in direction of the velocity vector and has an absolute magnitude given by

$$a_r = \frac{v^2}{r} \qquad \textbf{4.31}$$

where r is the radius of curvature of the path at the point under consideration.

The acceleration associated with uniform circular motion is called a **centripetal** (center seeking) acceleration.

During this acceleration at constant speed, the particle travels the distance $2\pi r$ (circumference) of the circle in time T given by

$$T = \frac{2\pi r}{v} \qquad \textbf{4.32}$$

which is called the period of revolution, or the period, of the motion. In general, T is the time for a particle to go round a closed path exactly once.

Circular Motion

Consider an object moving in a circle with a uniform speed round a fixed point O as centre as shown in Figure 4.10. Assume the object moves from A to B so that the radius OA moves through an angle θ. A particle at B is at a fixed distance r from the origin and rotates about it in a circle of radius r. We represent the position of B with its polar coordinates (r, θ), where r is the distance from the origin to B and θ is measured anticlockwise from some preferred direction like the positive x – direction.

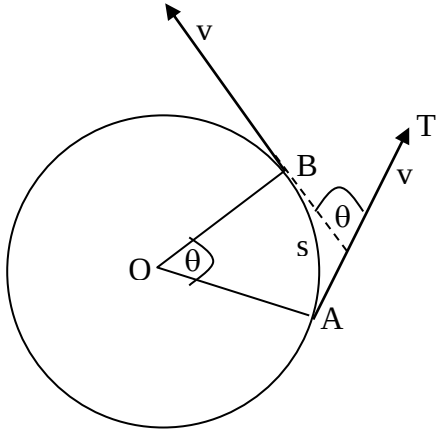


Figure 4.10 Circular motion

In this way only the angle θ changes with time; r remains constant. If s is the length of the arc AB, then s/r = θ, by definition of an angle in radians.

$$\therefore s = r\theta \qquad \textbf{4.33}$$

and

$$\theta = s/r \qquad \textbf{4.34}$$

One **radian** is *the angle subtended by an arc length equal to the radius of the arc*. Since the circumference of a circle is $2\pi r$, it follows from Equation 4.34 that 360^0 corresponds to an angle of $2\pi r/r$ rad $= 2\pi$ rad.

Hence $1rad = 360^0/2\pi \approx 57.3^0$. To convert an angle in degrees to an angle in radians, we have: $2\pi rad = 360^0$

$$\theta(rad) = \frac{\pi}{180^0}\theta(\deg) \qquad \textbf{4.35}$$

As an example, $60^0 = \pi/3\, rad$ and $90^0 = \pi/2\, rad$. As the object travels from position P to position Q in a time Δt as shown in Figure 4.11, the radius vector sweeps out an angle $\Delta\theta = \theta_f - \theta_i$ called the **angular displacement** of the body. That is

$$\Delta\theta = \theta_f - \theta_i$$

The angular speed, ω, about O is defined as the change of the angle per second. If t is the time taken to move from A to B as in Figure 4.10, then

$$\omega = \frac{\theta}{t} \qquad \textbf{4.36}$$

expressed in 'radians per second' (rad s^{-1}).

Define the average angular speed $\bar{\omega}$ as the ratio of the angular displacement to the time interval Δt (Figure 4.11)

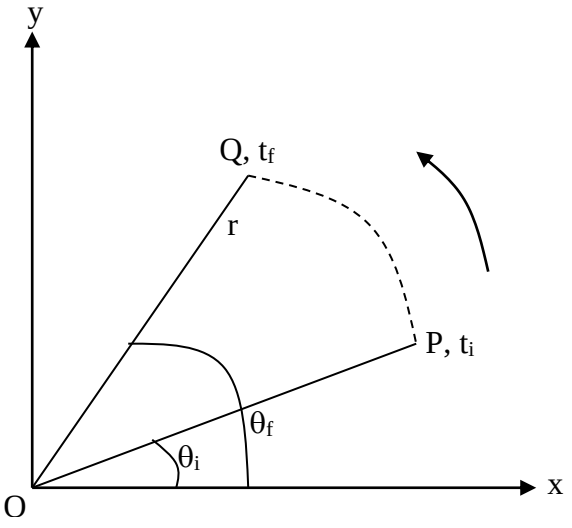


Figure 4.11 A particle on a rotating rigid object moves from P to Q along the arc of a circle. In the time interval $\Delta t = t_f - t_i$, the radius vector sweeps out an angle $\Delta\theta = \theta_f - \theta_i$.

$$\bar{\omega} = \frac{\theta_f - \theta_i}{t_f - t_i} = \frac{\Delta\theta}{\Delta t} \qquad \textbf{4.37}$$

The instantaneous angular speed ω is defined as the limit of the ratio $\Delta\theta/\Delta t$ as Δt goes to zero:

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \qquad \textbf{4.38}$$

ω is taken to be positive when θ is increasing (anticlockwise motion) and negative when θ is decreasing (clockwise motion).

Divide Equation 4.33 by t, the time to move from A to B in Figure 4.10,

$$\therefore \frac{s}{t} = r\frac{\theta}{t} \qquad \textbf{4.39}$$

$$\therefore v = r\omega \qquad \textbf{4.40}$$

If the angular velocity of a rotating body is not constant, then the body has an angular acceleration. Let ω_f and ω_i be the angular velocities at time t_f and t_i, respectively. The average angular acceleration of the rotating body in the interval from t_i to t_f (Δt) is defined as

$$\bar{\alpha} = \frac{\omega_f - \omega_i}{t_f - t_i} = \frac{\Delta\omega}{\Delta t} \qquad \textbf{4.41}$$

where $\Delta\omega$ is the change in the angular velocity that occurs during the time interval Δt. The instantaneous angular acceleration is defined as the limit of the ratio $\Delta\omega/\Delta t$ as Δt goes to zero:

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} \qquad \textbf{4.42}$$

The unit of angular acceleration is the radian per second-squared (rad/s^2) or the revolution per second –squared (rev/s^2).

Divide Equation 4.40 by t to get:

$$\frac{v}{t} = r\frac{\omega}{t} \qquad \textbf{4.43}$$

v/t is the tangential component a_t of the linear acceleration and ω/t is the angular acceleration α. Then

$$a_t = \alpha r \qquad \textbf{4.44}$$

As has been seen already, the radial component of the acceleration is:

$$a_r = \frac{v^2}{r} = \omega^2 r\,. \qquad \textbf{4.45}$$

Example 4.5 Express each of the following in terms of the other angular measures: (a) 28^0, (b) $\frac{1}{4} rev/s$, (c) 2.18 rad/s^2.

Solution

(a)

$$28^0 = (28\,deg)\left(\frac{1\,rev}{360\,deg}\right) = 0.078\,rev$$

$$= (28\,deg)\left(\frac{2\pi\,rad}{360\,deg}\right) = 0.49\,rad$$

(b)

$$\frac{1}{4}\frac{rev}{s} = \left(0.25\frac{rev}{s}\right)\left(\frac{360\,deg}{1\,rev}\right) = 90\frac{deg}{s}$$

$$= \left(0.25\frac{rev}{s}\right)\left(\frac{2\pi\,rad}{1\,rev}\right) = \frac{\pi}{2}\frac{rad}{s}$$

(c)

$$2.18\frac{rad}{s^2} = \left(2.18\frac{rad}{s^2}\right)\left(\frac{360\,deg}{2\pi\,rad}\right) = 125\frac{deg}{s^2}$$

$$= \left(2.18\frac{rad}{s^2}\right)\left(\frac{1\,rev}{2\pi\,rad}\right) = 0.35\frac{rev}{s^2}$$

Example 4.6 A car moves round a circular track of radius 0.3 m at 2 revolutions per second. What is

(a) the angular speed ω,

(b) the period T,

(c) the speed v of the car? Find also

(d) the angular speed of the car if it moves with a uniform speed of 2 ms^{-1} in a circle of radius 0.4 m.

Solution

(a) For 1 revolution, angle turned $\theta = 2\pi\, rad\left(360^0\right)$. So that

$\omega = 2 \times 2\pi = 4\pi\, rads^{-1}$

(b) Period $T =$ time for 1 rev. $= \dfrac{2\pi}{\omega} = \dfrac{2\pi}{4\pi} = 0.5\, s$

(c) Speed $v = r\omega = 0.3 \times 4\pi = 1.2\pi = 3.8\, ms^{-1}$

(d) From $v = r\omega$

$$\omega = \frac{v}{r} = \frac{2\, ms^{-1}}{0.4\, m} = 5\, rads^{-1}$$

Simple Harmonic Motion

Any motion that repeats itself at regular intervals is called **periodic motion** or **harmonic motion**. It turns out that for such motion the displacement x of the particle from the origin is given as a function of time.

Consider a physical system that consists of a block of mass *m* attached to the end of a spring, with the block free to move on a horizontal, frictionless surface as shown in Figure 4.12. When the spring is neither stretched nor compressed, the block is at the position $x = 0$, which is the equilibrium position of the system as shown in Figure 4.12b.

When the block is displaced a distance x, which is small, from equilibrium, the spring, with a spring constant k, exerts a force on the block which is proportional to the displacement and given by Hooke's law as

$$F = -kx \qquad \textbf{4.46}$$

The force is always directed towards the equilibrium position and therefore *opposite* the displacement and it is called a **restoring force**. This means that when the block is displaced to the left of $x = 0$ in Figure 4.12c then the displacement is negative and the restoring force is directed to the right. When the block is displaced to the right of $x = 0$, then the displacement is positive and the restoring force is directed to the left.

Let us apply Newton's second law to the motion of the block, together with Equation 4.46 to obtain

$$F = -kx = ma \qquad \textbf{4.47}$$

$$a = -\frac{k}{m}x \qquad \textbf{4.48}$$

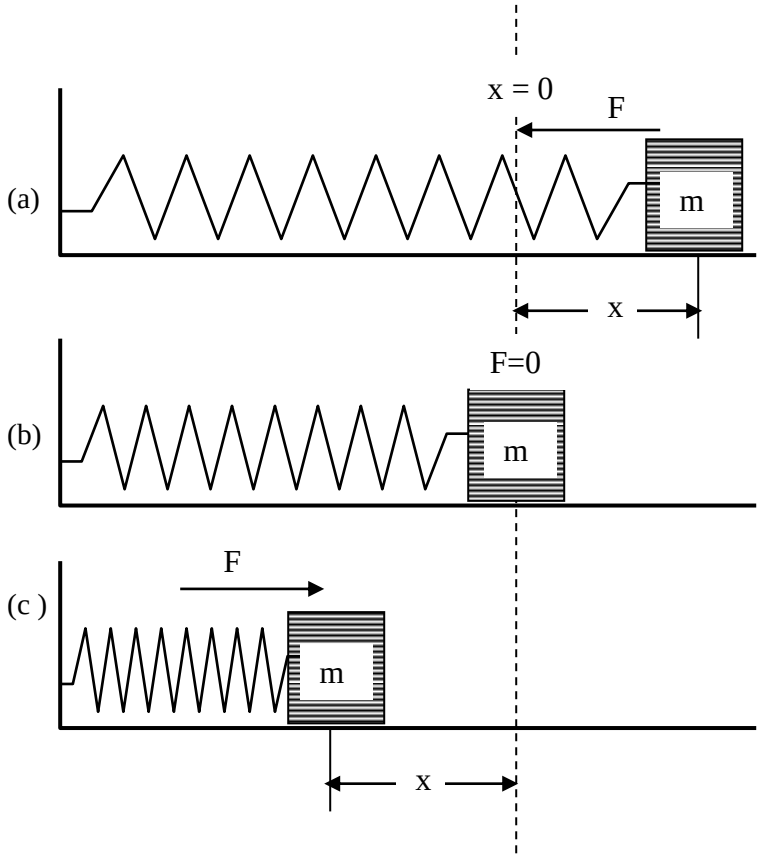


Figure 4.12 A block attached to a spring moving on a frictionless surface. (a) when the block is displaced to the right of equilibrium (x > 0), the force exerted by the spring acts to

But the acceleration is also given by

$$a = -\omega^2 x \qquad \textbf{4.49}$$

which means that

$k = m\omega^2$ and

$$\omega = \sqrt{k/m} \qquad \textbf{4.50}$$

From Equation 4.49 the acceleration is proportional to the displacement but opposite in sign, and the two quantities are related by the angular frequency-squared. Systems that behave this way are said to show **simple harmonic motion**. **Simple harmonic motion is the motion executed by a particle of mass m subject to a force that is proportional to the displacement of the particle but opposite in sign**.

In general, it turns out that for such motion the displacement x of the particle from the origin is given as a function of time by

$$x = A\cos(\omega t + \phi) \qquad \textbf{4.51}$$

where A, ω, and ϕ are constants. The quantity A, a positive constant whose value depends on how the motion was started, is called the **amplitude** of the motion. The cosine function in Equation 4.51 varies between the limits ± 1, so the displacement x varies between the limits $\pm A$, as shown in Figure 4.13.

The time-varying quantity $(\omega t + \phi)$ is called the **phase** of the motion, and the constant ϕ is called the **phase constant** (or **phase angle**). If the particle is at its maximum position $x = A$ at $t = 0$, then $\phi = 0$ as shown in Figure 4.13b. Let us interpret the constant ω. The displacement x must return to its initial value after

one period T of the motion. That is, x must equal $x(t+T)$ for all t. Let us put $\phi = 0$ in Equation 4.51 to simplify the analysis. From Equation 4.51 we have

$$A\cos\omega t = A\cos\left[\omega(t+T)\right] \qquad \textbf{4.52}$$

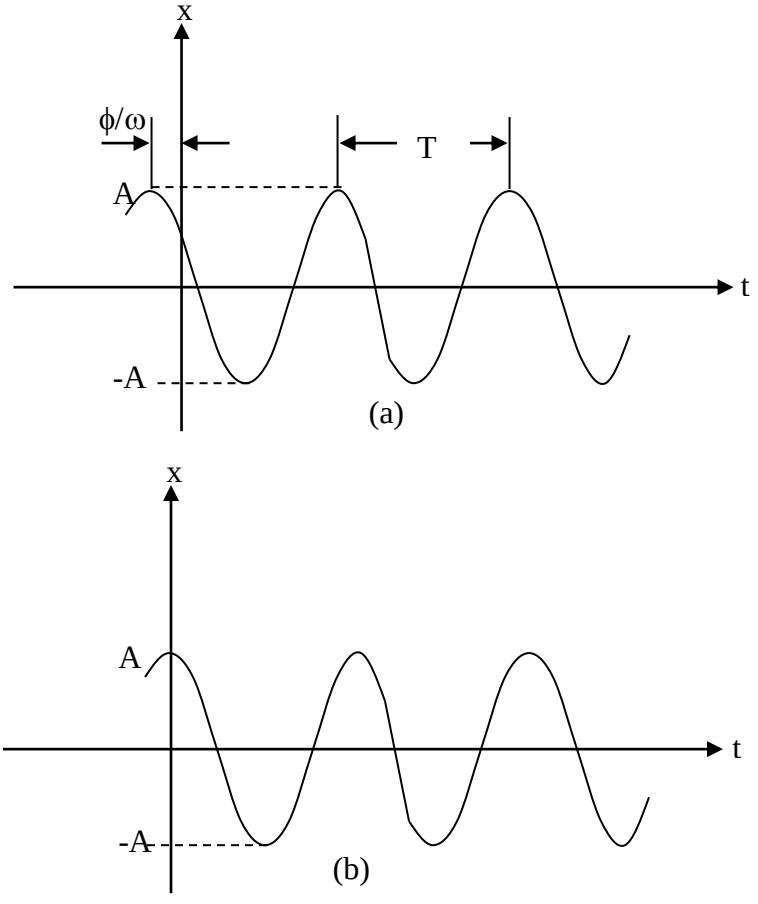


Figure 4.13 (a) An x – t curve for a particle undergoing simple harmonic motion. The amplitude of the motion is A, the period is T, and the phase constant is ϕ. (b) The x – t curve in the special case in which x = A at t = 0 and hence ϕ = 0.

The cosine function first repeats itself when its argument (the phase) has increased by 2π rad, so that we should have

$$\omega(t+T) = \omega t + 2\pi$$

$$\therefore \omega T = 2\pi \qquad \textbf{4.53}$$

Thus

$$\omega = \frac{2\pi}{T} \qquad \textbf{4.54}$$

The constant ω is called the angular frequency of the motion and has units of radians per second.

The linear velocity of a particle undergoing simple harmonic motion can be found by differentiating Equation 4.51 with respect to time:

$$v = \frac{dx}{dt} = \frac{d}{dt}\left[A\cos(\omega t + \phi)\right] \qquad \textbf{4.55}$$

which means

$$v = -\omega A\sin(\omega t + \phi) \qquad \textbf{4.56}$$

From Equation 4.56, the acceleration of the particle is

$$a = \frac{dv}{dt} = \frac{d}{dt}\left[-\omega A\sin(\omega t + \phi)\right]$$

which gives

$$a = -\omega^2 A\cos(\omega t + \phi) \qquad \textbf{4.57}$$

But $x = A\cos(\omega t + \phi)$ so Equation 4.57 becomes

$$a = -\omega^2 x \qquad \textbf{4.58}$$

From Equation 4.57, because the sine function oscillates between ± 1, the extreme values of v are $\pm \omega A$. From Equation 4.58, and since the cosine function oscillates between ± 1, the extreme values of a are $\pm \omega^2 A$. Therefore, the maximum speed and the magnitude of the maximum acceleration of a particle moving in simple harmonic motion are

$$v_{\max} = \omega A \qquad \textbf{4.59}$$

$$a_{\max} = \omega^2 A \qquad \textbf{4.60}$$

The importance of the phase constant can be seen when comparing the motion of two or more oscillating objects. Let us imagine two identical pendulum bobs swinging side by side in simple harmonic motion, with one being released later than the other. Then the bobs have different phase constants. We can also show how the phase constant and the amplitude of any particle moving in simple harmonic motion can be determined if we know the particle's initial speed and position and the angular frequency of its motion.

Now suppose that at $t = 0$ the initial position of a single oscillator is $x = x_i$ and its initial speed is $v = v_i$. Under such conditions, Equation 4.51 and Equation 4.56 give

$$x_i = A\cos\phi \qquad \textbf{4.61}$$

$$v_i = -\omega A\sin\phi \qquad \textbf{4.62}$$

Equation 4.62 divided by Equation 4.61 eliminates A, to give $v_i / x_i = -\omega\tan\phi$, or

$$\tan\phi = \frac{-v_i}{\omega x_i} \qquad \textbf{4.63}$$

Square Equations 4.61 and 4.62, divide Equation 4.62 by ω^2, and then add terms to obtain,

$$x_i^2 + \left(\frac{v_i}{\omega}\right)^2 = A^2\cos^2\phi + A^2\sin^2\phi .$$

But $\sin^2 + \cos^2 = 1$, then

$$A = \sqrt{x_i^2 + \left(\frac{v_i}{\omega}\right)^2} \qquad \textbf{4.64}$$

Recall that the period of any simple harmonic oscillator is $T = 2\pi/\omega$ and that the frequency is the inverse of the period. Also $\omega = \sqrt{k/m}$, so the period and the frequency of the spring-block system become

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \qquad \textbf{4.65}$$

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad \textbf{4.66}$$

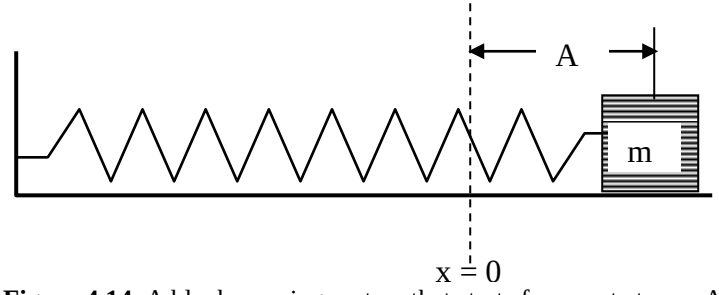
Equations 4.65 and 4.66 tell us that a large angular frequency and thus a small period go with a stiff spring (i.e. large k) and a light block (small m). Furthermore, the frequency and period are independent of the amplitude of the motion.

Energy in Simple Harmonic Motion

We examine the mechanical energy of the block-spring system illustrated in Figure. 4.14. Since the surface is assumed frictionless, it is expected that the total mechanical energy will be constant. The potential energy of a linear oscillator like that of Figure 4.14 is associated entirely with the spring. Its value depends on how much the spring is stretched or compressed, that is, on x.

The elastic potential energy stored in the spring of any elongation x is given by $\frac{1}{2}kx^2$. From Equation 4.51, we have

$$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega t + \phi) \qquad \textbf{4.67}$$



t = 0
x_i = A. x = A cos ωt
v_i = 0

Figure 4.14 A block – spring system that starts from rest at x_i = A. In this case ϕ = 0 and thus x = A cos ωt.

The kinetic energy of the system is associated with the block only. Its value depends on how fast the block is moving, that is, on v. From Equation 4.56, the kinetic energy can be expressed as

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2A^2\sin^2(\omega t + \phi) \qquad \textbf{4.68}$$

U and K are always positive quantities. Since $\omega^2 = k/m$, the mechanical energy of the simple harmonic oscillator can be expressed as

$$E = U + K = \frac{1}{2}kA^2\left[\sin^2(\omega t + \phi) + \cos^2(\omega t + \phi)\right] \qquad \textbf{4.69}$$

But the identity $\sin^2\theta + \cos^2\theta = 1$, so that if we let $\theta = \omega t + \phi$ for the quantity in bracket, then

$$E = \frac{1}{2}kA^2 \qquad \textbf{4.70}$$

The mechanical energy of a linear oscillator is indeed constant, independent of time. **The total mechanical energy of a simple harmonic oscillator is a constant of the motion and is proportional to the square of the amplitude.** Note also that U is small when K is large, and vice versa, because the sum must be constant.

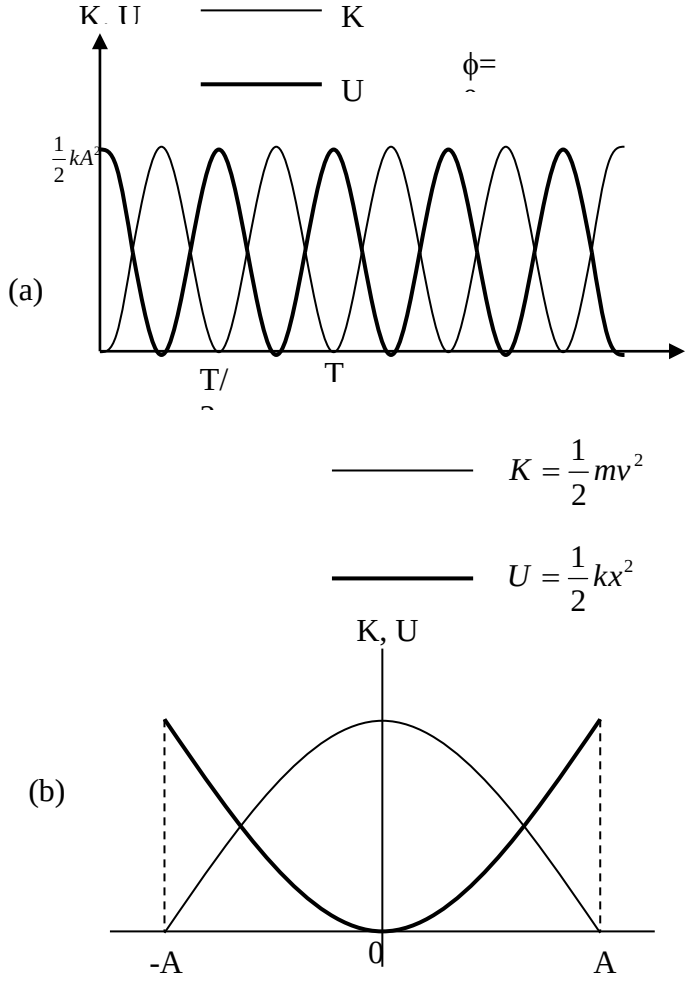


Figure 4.15 (a) Kinetic energy and potential energy verses time for a simple harmonic oscillator with $\phi = 0$. (b) Kinetic energy and potential energy verses displacement for a simple harmonic oscillator. In either

The total mechanical energy is equal to the maximum potential energy stored in the spring when $x = \pm A$ because $v = 0$ at these points and thus there is no kinetic energy. At the equilibrium position, where $U = 0$ because $x = 0$, the total energy, all in the form of kinetic energy, is again $\frac{1}{2}kA^2$ .That is

$$E = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}m\frac{k}{m}A^2 = \frac{1}{2}kA^2 \qquad \textbf{4.71}$$

when $x = 0$.

Plots of the kinetic and potential energies against time are shown in Figure. 4.15a, with $\phi = 0$. The variations of K and U with the displacement x of the block are plotted in Figure 4.15b. Energy is continuously being transformed between potential energy stored in the spring and kinetic energy of the block.

It can now be understood why an oscillating system contains an element of springiness and an element of inertia: it uses the former to store its potential energy and the latter to store its kinetic energy.

Example 4.7 A spring makes 12 vibrations in 40 s. Find the period and frequency of the vibration.

Solution

$$T = \frac{elapsed\ time}{vibrations\ made} = \frac{40s}{12} = 3.3s$$

$$f = \frac{\text{vibrations made}}{\text{elapsed time}} = \frac{12}{40s} = 0.30Hz$$

Example 4.8 When a 400 g mass is hung at the end of a vertical light spring, the spring stretches 35 cm. What is the spring constant, k, and how much further will it stretch if an additional 400 g mass is hung from it?

Solution

We know that

$F = ky$, since the spring is vertical, where

$F = mg = (0.400kg)(9.8m/s^2) = 3.92N$ so that

$$k = \frac{F}{y} = \frac{3.92N}{0.35m} = 11.2\,N/m$$

With an additional 400 g load, the total force stretching the spring is 2F = 7.84 N. Then

$$y = \frac{F}{k} = \frac{7.84N}{11.2\,N/m} = 0.70m = 2 \times 35cm$$

So that each 400 g load stretches the spring by the same amount whether the spring is loaded or not.

Example 4.9 A 200 g mass vibrates horizontally without friction at the end of a horizontal spring for which k = 7.0 N/m. The mass is displaced 5.0 cm from equilibrium and released. Find (a) its maximum speed and (b) its speed when it is 3.0 cm from equilibrium. (c) What is its acceleration in each of these cases?

Solution

Using the conservation of energy law in this case gives,

$\frac{1}{2}x_0^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$, where

$k = 7.0\,N/m, x_0 = 0.050m,$ and $m = 0.200kg.$ Then v becomes

$$v = \sqrt{\frac{k}{m}\left(x_0^2 - x^2\right)}$$

(a) For maximum speed $x = 0$; i.e., when the mass is passing through the equilibrium position:

$$v = x_0\sqrt{\frac{k}{m}} = \left(0.05m\right)\sqrt{\frac{7.0\,N/m}{0.200kg}} = 0.296\,m/s$$

(b) When $x = 0.030m$,

$$v = \sqrt{\frac{7.0\,N/m}{0.200kg}\left[\left(0.050\right)^2 - \left(0.030\right)^2\right]m^2} = 0.237\,m/s$$

(c) Using $F = ma$ and $F = kx$, we have

$a = \frac{k}{m}x = \left(35s^{-2}\right)\left(x\right)$ which gives a = 0 when the mass is at $x = 0$ and

$a = 1.05\,m/s^2$ when $x = 0.030m$

Example 4.10 A 50 g mass vibrates in S.H.M at the end of a light spring. The amplitude of the motion is 12 cm, and the period is 1.70 s. Find: (a) the frequency, (b) the spring constant, (c) the maximum speed of the mass, (d) the maximum acceleration of the mass, (e) the speed when the displacement is 6 cm, and (f) the acceleration when $x = 6\,cm$.

Solution

(a) $f = \frac{1}{T} = \frac{1}{1.70\,s} = 0.588\,H_z$

(b) We know that $T = 2\pi\sqrt{m/k}$, so that

$$k = \frac{4\pi^2 m}{T^2} = \frac{4\pi^2\left(0.050kg\right)}{\left(1.70s\right)^2} = 0.68\,N/m$$

(c) $v_0 = x_0\sqrt{\frac{k}{m}} = \left(0.12m\right)\sqrt{\frac{0.68\,N/m}{0.050kg}} = 0.44\,m/s$

(d) From $a = -\left(k/m\right)x$, a has a maximum magnitude when x is maximum, i.e., at the endpoints $x = \pm x_0$. So that

$$a_0 = \frac{k}{m}x_0 = \frac{0.68\,N/m}{0.050kg}\left(0.12m\right) = 1.63\,m/s^2$$

(e) We know $|v| = \sqrt{\left(x_0^2 - x^2\right)}\left(k/m\right)$,

$$|v| = \sqrt{\left(0.12^2 - 0.06^2\right)\frac{0.68}{0.050}} = 0.38\, m/s$$

(f) $a = -\frac{k}{m}x = -\frac{0.68\, N/m}{0.050 kg}\left(0.06m\right) = -0.82\, m/s^2$

Example 4.11 A 20 kg electric motor is mounted on four vertical springs, each having a spring constant of $30\, N/cm$. Find the period with which the motor vibrates vertically.

Solution

The four springs can be replaced by an equivalent spring an its spring constant becomes $4\left(3000\, N/m\right)$ or $12000\, N/m$. Then

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{20kg}{12000\, N/m}} = 0.256\, s$$

Pendulums

Let us turn now to a class of simple harmonic oscillators in which the springiness is associated with the gravitational force rather than with the elastic properties of a twisted wire or a compressed or stretched spring.

The Simple Pendulum

It is another mechanical system that exhibits periodic motion. It consists of a particle-like bob of mass m suspended from an unstretchable, massless string of length L fixed at the upper end as shown Figure 4.16a . The bob is free to swing back and forth in the plane of the page, to the left and right of a vertical line through the point at which the upper end of the string is fixed. The forces acting on the particle, shown in Figure 4.16b are its weight m**g** and the tension **T** in the string. The weight can be resolved into a radial component $mg\cos\theta$ and a component $mg\sin\theta$ tangent to the path taken by the particle. This tangential component is a restoring force, since it always acts opposite the displacement of the particle so as to bring the particle back toward its central location, the equilibrium position ($\theta = 0$), where it would be at rest were it not swinging.

We use Newton's second law to write the restoring force as

$$F_t = -mg\sin\theta = m\frac{d^2s}{dt^2} \qquad \textbf{4.72}$$

where s is the bob's displacement measured along the arc and the minus sign indicates that the tangential force acts toward the equilibrium (vertical) position. Since $s = L\theta$ and L is constant, Equation 4.72 reduces to

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta \qquad \textbf{4.73}$$

For small angles (i.e. $\sin\theta \approx \theta$) the equation of motion for the simple pendulum becomes

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta \qquad \textbf{4.74}$$

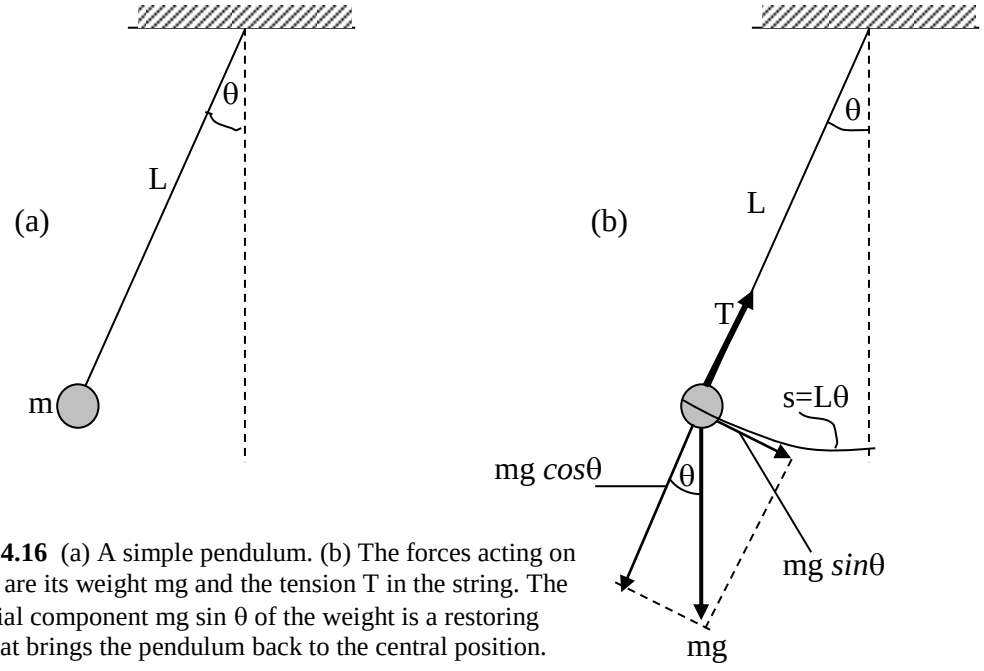


Figure 4.16 (a) A simple pendulum. (b) The forces acting on the bob are its weight mg and the tension T in the string. The tangential component mg sin θ of the weight is a restoring force that brings the pendulum back to the central position.

Equation 4.74 is in the form of Equation 4.58, and we conclude that the motion for small amplitudes of oscillation is a simple harmonic motion. Therefore, θ can be written as $\theta = \theta_{\max} \cos(\omega t + \phi)$, where $\theta_{\max}$ is the maximum angular displacement and the angular frequency ω is

$$\omega = \sqrt{\frac{g}{L}} \qquad \textbf{4.75}$$

The period of the motion of a simple pendulum is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}\,. \qquad \textbf{4.76}$$

We see that, the period and frequency of a simple pendulum depend only on the length of the string and the acceleration due to gravity. Because the period is

independent of the mass, we conclude that all simple pendulums that are of equal length and are at the same location (so that g is constant) oscillate with the same period.

The conical Pendulum

Suppose a small object of mass m is tied to a string OA of length l and then whirled round in a horizontal circle of radius r, with O fixed directly above the centre B of the circle as shown in Figure 4.17. If the circular speed of A is constant, the string turns at a constant angle θ to the vertical. This is called a conical pendulum.

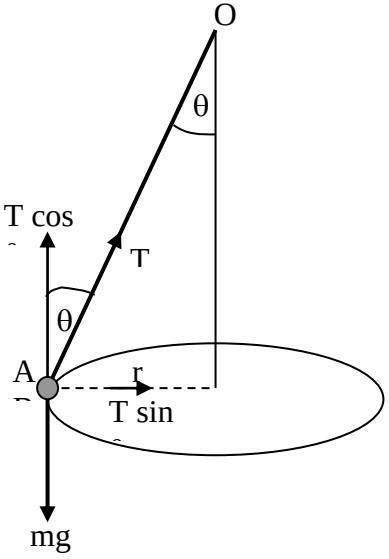


Figure 4.17 Conical pendulum

Since A moves with a constant speed **v** in a circle of radius r, there must be a centripetal force, mv^2/r, acting towards the centre B. The horizontal component, $T\sin\theta$, of the tension in the string provides this force along AB. So

$$T\sin\theta = \frac{mv^2}{r} \qquad \textbf{4.77}$$

Also, since the mass does not move vertically, its weight mg must be counterbalanced by the vertical component $T\cos\theta$ of the tension. So

$$T\cos\theta = mg \qquad \textbf{4.78}$$

Dividing Equation 4.77 by Equation 4.78 gives

$$\tan\theta = \frac{v^2}{rg} \qquad \textbf{4.79}$$

If a parallel beam of light is used to cast a shadow of the bub A on a screen, the motion of the shadow will be simple harmonic, maximum displacement is r, and $\omega = v/r$.

Example 4.12. A pendulum bob has a mass of 0.5kg. It is suspended by a cord 2 m long which is pulled back through an angle of 36.9^0 and released. Find (a) its maximum potential energy relative to its lowest position, (b) its potential energy when the cord makes an angle of 10^0 with the vertical, (c) its maximum speed, and (d) its speed when the cord makes an angle of 10^0 with the vertical.

Solution

$h = 2 - 2cos36.9^0 = 2 - 1.6 = 0.4\text{m}$

(a) PE_{max} = (0.5 x 9.8)0.4 = 1.96 J

(b) When $\theta = 10^0,\ h_{10} = 2 - 2cos10^0 = 2(1 - 0.9848) = 0.0304\text{m}$

$PE_{10^0} = (0.5 \times 9.8) \times 0.0304 = 0.149\text{J}$

(c) At bottom $\frac{1}{2}\left(\frac{1}{2}\right)v_{max}^2 = 1.96 \qquad v_{max} = 2 \times 1.4 = 2.8\,m/s$

(d) $(1.96 - 0.149)\text{J} = \frac{1}{2}\left(\frac{1}{2}\right)v_{10}^2$

$v_{10}^2 = 7.24 \qquad v_{10} = 2.69\,\text{m/s}$

Projectile Motion

Anyone observing the motion of a stone or any object thrown into the air is observing projectile motion. The stone moves in a curved path, and the motion is simple to analyze if two assumptions are made: (1) the acceleration of free-fall **g** is constant over the range of motion and is directed downward, and (2) the effect of air resistance is negligible. Using these assumptions, it is found that the path of a projectile, also called *trajectory*, is always a parabola. Projectile motion looks complicated but we have the following simplifying feature:

The horizontal motion and vertical motion are independent of each other; that is, neither motion affects the other.

To show that the trajectory of a projectile is a parabola, we choose our reference frame such that the y direction is vertical and positive is upward. Neglecting air resistance, we have $a_y = -g$ and that $a_x = 0$ (the projectile has *no* horizontal acceleration). Furthermore, let us assume that at $t = 0$, the projectile leaves the origin ($x_0 = y_0 = 0$) with speed v_0 as shown in Figure 4.18.

The vector $\mathbf{v_0}$ makes an angle θ_0 with the horizontal, where θ_0 is the angle at which the projectile leaves the origin. The initial velocity $\mathbf{v_0}$ can be written as

$$\mathbf{v}_0 = v_{0x}\mathbf{i} + v_{0y}\mathbf{j} \qquad \textbf{4.80}$$

The components v_{0x} and v_{0y} can be found if the angle between $\mathbf{v_0}$ and the positive x direction θ_0 is known:

$$v_{0x} = v_0 \cos\theta_0 \quad \text{and} \quad v_{0y} = v_o \sin\theta_0 \qquad \textbf{4.81}$$

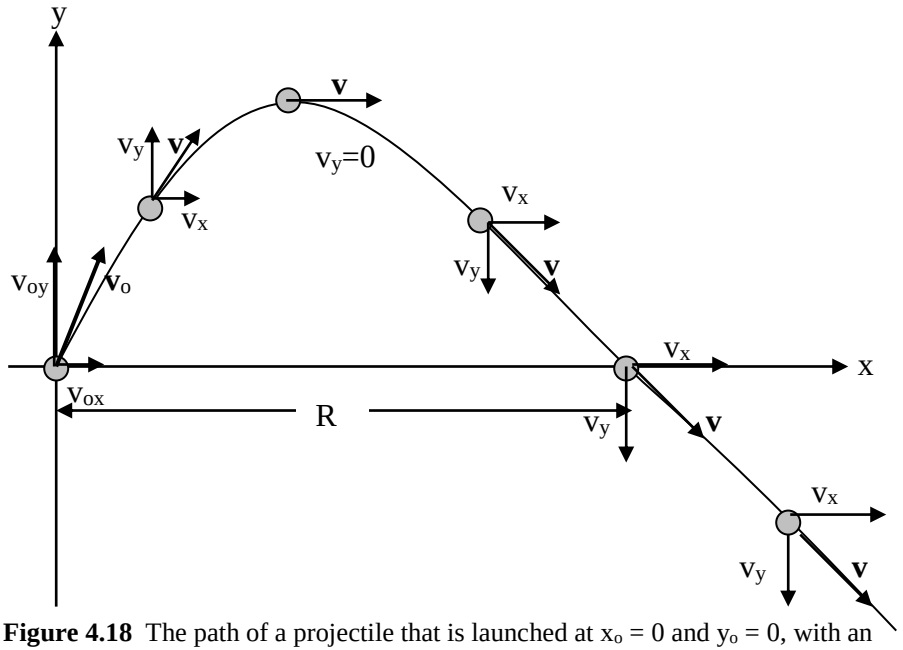


Figure 4.18 The path of a projectile that is launched at $x_o = 0$ and $y_o = 0$, with an initial velocity v_o. The initial velocity and the velocities at various points along its path are shown, along with their components. Note that the horizontal velocity component remains constant but the vertical velocity component changes continuously. The range R is the horizontal distance the projectile has traveled when it returns to its launch height.

As stated earlier for projectile motion, the horizontal motion and vertical motion are independent of each other; that is neither motion has an effect on the other. This allows breaking up a problem involving two-dimensional motion into two separate and easier one-dimensional problems, one for the horizontal motion and one for the vertical motion.

Analysis of projectile motion

The Horizontal motion

Since there is no acceleration in the horizontal direction, the horizontal component $\mathbf{v}_{ox}$ of the projectile's initial velocity remains unchanged throughout the motion. The horizontal displacement $x - x_0$ from an initial position x_o is given as

$$x - x_0 = v_{0x}t \qquad \textbf{4.82}$$

since $a = 0$.

But $v_{0x} = v_0 \cos\theta_0$, so that

$$x - x_0 = (v_0 \cos\theta_0)t \qquad \textbf{4.83}$$

The vertical motion

The vertical motion is the motion for a particle in free fall, i.e.

$$y - y_0 = v_{0y}t - \frac{1}{2}gt^2$$

$$= (v_0 \sin\theta_0)t - \frac{1}{2}gt^2 \qquad \textbf{4.84}$$

where the initial vertical velocity component $\mathbf{v}_{0y}$ is replaced with the equivalent $\mathbf{v}_o \sin\theta_o$.

The equation of the path

To find the equation of the path (the trajectory) of the projectile, eliminate t between Equations 4.83 and 4.84. Solve Equation 4.83 for t and substitute into Equation 4.84 and obtain after a little rearrangement,

$$y = (\tan\theta_0)x - \frac{gx^2}{2(v_0\cos\theta_0)^2} \qquad \textbf{4.85}$$

This is the equation of the path shown in Figure 4.18.

For simplicity in the derivation, we let $x_o = 0$ and $y_o = 0$ in Equations 4.83 and 4.84, respectively. Since g, θ_o, and v_o are constants, Equation 4.85 is of the form $y = ax + bx^2$, in which a and b are constants. This is the equation of a parabola, so the path is parabolic.

The Horizontal Range

The horizontal range R of the projectile is the horizontal distance the projectile has traveled when it returns to its initial (launch) height as shown in Figure 4.18. To find the range R of the projectile, put $x - x_0 = R$ in Equation 4.83 and $y - y_0 = 0$ in Equation 4.84, obtaining

$$x - x_0 = (v_0\cos\theta_0)t = R \qquad \textbf{4.86}$$

and

$$y - y_0 = (v_0\sin\theta_0)t - \frac{1}{2}gt^2 = 0 \qquad \textbf{4.87}$$

Eliminating t between Equations 4.86 and 4.87 yields

$$R = \frac{2v_0^2}{g}\sin\theta_0\cos\theta_0 \qquad \textbf{4.88}$$

Using the identity $\sin 2\theta_0 = 2\sin\theta_0\cos\theta_0$ we obtain

$$R = \frac{v_0^2}{g}\sin 2\theta_0 \qquad \textbf{4.89}$$

Note that R has a maximum value when $\sin 2\theta_0 = 1$, which corresponds to $2\theta_o = 90$ or $\theta_o = 45^0$.

The horizontal range R is a maximum for a launch angle of 45^0.

We have assumed throughout that the air through which the projectile moves has no effect on its motion, which is a reasonable assumption at low speeds. However, at higher speeds, the disagreement between our calculations and the actual motion of the projectile can be large because of air resistance offered to the motion.

Example 4.13. A marble with speed 20 cm/s rolls off the edge of a table 80 cm high. How long does it take to drop to the floor? How far, horizontally, from the table edge does the marble strike the floor?

Solution

Choose downward as positive with origin at edge of table top.

$v_{ox} = v_0 = 20\,\text{cm/s} \quad v_{oy} = 0 \quad a_y = +g = +980\,\text{cm/s}^2 \quad a_x = 0$

To find time of fall, $y = v_{oy}t + \frac{1}{2}gt^2$, or 80 cm$= 0 + (490\,\text{cm/s}^2)t^2$; t = 0.40 s.

The horizontal distance is gotten from $x = v_{ox}t = (20\,\text{cm/s})(0.40\text{s}) = 8.0\text{cm}$.

Example 4.14. A marble traveling at 100 cm/s rolls off the edge of a level table. If it hits the floor 30 cm away from the spot directly below the edge of the table, how high is the table?

Solution

This is a projectile problem with v_{ox} = 100 cm/s. For the horizontal motion:

$s_x = v_{ox}t \qquad 30 = 100t \qquad t = 0.30\ \text{s}$

For the vertical motion:

$$s_y = v_{oy}t + \frac{1}{2}at^2 = 0 + \frac{1}{2}(980)(0.30)^2 = 44.1\text{cm}$$

Example 4.15. A golf ball leaves the golf club at an angle of 60^0 above the horizontal with a velocity of 30 m/s. (a) How high does it go? (b) Assuming a level fairway, determine how far away it hits the ground.

Solution

(a) Taking up as positive, $v_{0y} = 30sin60^0 = 15\sqrt{3}\,\text{m/s}$ and, at maximum height, v_y equals zero. Thus,

$$v_y^2 = v_{oy}^2 + 2ah \qquad 0 = 675 - 19.6h \qquad h = \frac{675}{19.6} = 34.4\text{m}$$

(b) $x = \frac{v_0^2 sin2\theta}{g} = \frac{900(\sqrt{3}/2)}{9.8} = 79.5\text{m}$